Generative AI Audit Log — PS 6

CIS 731 — Fall 2026 — Semantic Search, Chunking & VDB Schema

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Assignment	PS 6: Semantic Search, Chunking, & VDB Schema
Schema authority	DERIVED (CIS 731, INV-11) — the instructor ontology is not reused
Tool audited	Claude (Anthropic) — conversational assistant
Period covered	27 October 2026
Deliverables	PS6_Abdulla_Mansoor.pdf; schema/ under Git tag ps6-submission

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1. Scope

Section 2 of the PS 6 specification directs students to use the Generative AI Kit across five stages, each paired with a verification duty: draft the cost model *then verify the arithmetic independently*; tabulate the index trade-offs *then check every claim against FAISS documentation*; generate the JSON Schema *then validate it with a validator recorded in the lockfile*; draft the contracts; and run an adversarial review for provenance holes. AI use here is instructed, not merely tolerated. This log records what the assistant did, what I verified, and — at greater length — what verification actually caught.

2. Assistance by specification stage

Stage	Assistant contribution	My verification
1 — Cost model	Drafted the fixed-window / sentence-boundary / structure-aware derivations and the storage formula.	Checked that 1/(1−p) and s/(s−o) agree, which is the arithmetic identity the spec i
2 — Index matrix	Wrote the benchmark harness over IndexFlatIP, IndexIVFFlat and IndexHNSWFlat.	Required the table be MEASURED rather than quoted. This is what exposed fin
3 — Schema synthesis	Generated the Draft 2020-12 schema including the if/then/else conditionals.	Validated with jsonschema 4.26.0, pinned in requirements.lock. Added 13 delib
4 — Contract drafting	Drafted admit_document and vdb_upsert with deal decorators.	Found finding 3.2 — the contracts were not running at all. Added negative test
5 — Adversarial review	Asked for provenance holes: orphan vectors, unlicensed documents, duplicate chunk_ids.	Findings 3.3–3.5 below. The licence hole is a real compliance gap in the team's

3. Findings — defects caught by running the work, not reading it

3.1 The index benchmark was measuring nothing (methodology error, mine)

The first benchmark run reported 1–30% recall@10 for *both* IVF and HNSW at every parameter setting. The reflex reading was that HNSW handles inner product badly. That was wrong. The synthetic corpus added per-coordinate Gaussian noise of 0.35 to unit-norm cluster centres; in 384 dimensions that perturbation has expected norm $0.35 \cdot \sqrt{384} \approx 6.9$, so after normalization the points were uniform on the sphere with no cluster structure at all. Near-zero ANN recall is the *correct* result for uniform high-dimensional data and a meaningless one for an embedding space. Scaling the noise by $1/\sqrt{d}$ fixed it: IVF went to 0.9999 at nlist=256/nprobe=4 and HNSW to 0.9802 at M=32/ef=64. Had I accepted the first table, the report would have recommended exact search for a confidently stated and entirely false reason.

3.2 The deal contracts were not being evaluated

admit_document's preconditions were written as single-parameter lambdas on a two-parameter function. A deal validator is invoked with the guarded function's full argument list, so each one raised `TypeError: <lambda>() takes 1 positional argument but 2 were given` instead of evaluating. The function still raised, so a careless harness that only asserted 'bad input raises' would have passed — the contract would have appeared to work while checking nothing. The negative tests assert the specific exception type (deal.PreContractError), which is what made the difference visible.

3.3 Duplicate chunk_id: IndexIDMap2.add_with_ids does not replace

The adversarial prompt asked where a duplicate chunk_id could arise. Measured answer: calling add_with_ids with an id already present does not overwrite it — ntotal went from 20,000 to 20,001 and the index held two vectors under one key, with a search free to return the stale one. A re-embedding run would have silently doubled part of the index. Two fixes, both now specified: chunk_id is derived deterministically as SHA-256(doc_id|chunk_index|chunker_version) so re-chunking produces the same key rather than a fresh one; and update is remove_ids followed by add_with_ids, which also sweeps up any pre-existing duplicate because remove_ids deletes every copy.

3.4 Orphan vectors: deletion order and the sidecar map

Two orphan paths were identified. First, deleting the record before the vector leaves a window in which a vector is retrievable with no record behind it; deletion is now ordered vector-first, so a crash mid-delete yields a missing embedding — detectable by EmbeddingCompleteness and repairable by re-embedding — instead of an unattributable vector that no scan would notice. Second, removing an id from FAISS does not remove it from the int64→chunk_id sidecar; the demo asserts both sides return to bijection after a delete. NoOrphanEmbeddings and EmbeddingCompleteness are checked as a pair precisely because each alone permits one of these states.

3.5 Unlicensed documents: an INV-9 gap in the team's own plan

The review asked which documents could enter the corpus without a licence check. The answer was not a code bug: INV-9 restricts corpus content to public-domain or instructor-provided material, while the team's acquisition plan of record from PS 4 is the authenticated Genius REST API — Tier 2, not public domain. I wrote admit_document to the invariant rather than to the plan: licensed_api and fair_use_excerpt are refused, and the harness asserts both refusals. Those values remain in the schema enum on purpose, so non-conforming material can be named and rejected rather than quietly relabelled as something admissible. The consequence is stated plainly in the report: the corpus the pipeline is currently specified against would be rejected by its own ingest gate.

3.6 The PDF silently corrupted its own mathematics

The first typeset build used ReportLab's built-in Helvetica, which has no glyphs for the notation the report depends on. U+27E8/27E9 (angle brackets) and U+2308/2309 (ceiling) both fell back to a double-bar glyph, so the inner-product proof rendered as though it were about norms — $\|u,v\|$ where $\langle u,v\rangle$ was written — and the chunking formula appeared to take a norm of a quotient. No error, no warning, and the corruption landed in exactly the two places where notation carries the argument. Caught by rendering the PDF to an image and reading it rather than trusting that the build succeeded. Fixed by registering DejaVu Sans.

4. What AI was not used for

No measured value was generated, estimated or interpolated by AI: every number in Part 1 is emitted by code in this repository and stored under out/, and ./run_all.sh regenerates all of it. The 36 validation cases are executed, not described. Where a quantity could not be measured in this environment — mean tokens per document, and the encoder and cross-encoder latencies, because the model files could not be downloaded — it is labelled an estimate at the point of use in the report rather than presented as a result. No citation was invented. The ontology's class hierarchy, relation cardinalities and competency questions follow my PS 4 Task 5 design; the assistant formalised and stress-tested them.

5. Attestation

I can defend each decision in the report without assistance: why the granularity ceiling binds before the model ceiling, why cosine and inner product coincide only on the unit sphere and what that implies for the unit-norm precondition, why Recall@k is not gradable on own data before INV-17, why attestedIn carries minimum cardinality 1, and why exact search is the right choice at this corpus scale despite the ANN analysis. Where verification changed a conclusion, §3 says so rather than presenting the corrected version as though it had been the first one.

Signed: ______________________________ Date: ______________